MTH1002

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

January 2018

Mathematical Methods

Module Leader: Dr Pascal Philipp

Duration: 2 HOURS.

The mark for this module is calculated from 30% of the percentage mark for this paper plus 70% of the percentage mark for the May 2018 paper.

Answer ALL questions.

Marks shown in questions are merely a guideline.

Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED BOOK** examination.

1. (a) Solve the equation $7\sinh(x) + 3\cosh(x) = 9.5$. (4)

(b) Consider the function $f(x) = e^{x^2} + 1$. Write down the maximal subset of $x \in \mathbb{R}$ that could be

(i) the domain,

(ii) the range of $f(x)$.

Explain why $f(x)$ has an inverse on $x \in [0, 1]$ but not on $x \in [-1, 1]$. By restricting $f(x)$ to the domain $x \in [0, 1]$ find an inverse of $f(x)$. (5)

(c) Consider the parabola $y^2 = 4(x + 1)$.

(i) Show that the equations $x(t) = t^2 - 1$ and $y(t) = 2t$ are a parametrization of this parabola.

(ii) Find the equation of the tangent to the parabola at a general point $(x(t), y(t))$.

(iii) Write down the equation of the tangent at the point P corresponding to $t = 1$, and show this on a sketch together with the parabola. (11)

[20]

2. (a) Find a formula for the general term of each of the following sequences

(i) $0, \frac{1}{2^2}, \frac{2}{3^2}, \frac{3}{4^2}, \frac{4}{5^2}, \ldots$

(ii) $\left(1 - \frac{1}{2}\right), \left(\frac{1}{3} - \frac{1}{2}\right), \left(\frac{1}{3} - \frac{1}{4}\right), \left(\frac{1}{5} - \frac{1}{4}\right), \left(\frac{1}{5} - \frac{1}{6}\right), \ldots$ (5)

(b) Consider the sequence given by the recursion formula

$$a_{n+1} = \frac{1}{2}\left(a_n + \frac{2}{a_n}\right)$$

with $a_0 = 1$.

(i) Write down the values of the first three terms of this sequence.

(ii) Assuming that the sequence converges, show that its limit is $\sqrt{2}$. (5)

(c) Find the following limits (if they exist), being careful to justify your answers

(i) $\lim_{n\to\infty} \dfrac{4n^2-1}{n^2+\sqrt{n}}$,

(ii) $\lim_{n\to\infty} \sqrt[n]{3^n+n+1}$,

(iii) $\lim_{x\to 0} \dfrac{e^x-x-1}{(\sin(x))^2}$. (10)

[20]

3. (a) Investigate the convergence or divergence of the following series, carefully justifying your reasons,

(i) $\displaystyle\sum_{k=1}^{\infty} \frac{7}{3k}$,

(ii) $\displaystyle\sum_{k=1}^{\infty} (-1)^k \frac{\ln(k)}{1+\sqrt{k}}$. (7)

(b) Show that the series

$$\sum_{k=1}^{\infty} \frac{1}{(k+4)(k+5)}$$

is a telescoping series, and find its sum. (5)

(c) Suppose that a function $f(x)$ is represented by the power series

$$f(x) = 1 - \frac{x-5}{4} + \frac{(x-5)^2}{4^2} - \frac{(x-5)^3}{4^3} + \ldots$$

(i) Find the values of x for which the series converges.

(ii) Find $f(4)$. (8)

[20]

4. (a) Determine whether the following statements are true or false. In each case give a proof or counterexample.

(i) If $f(x)$ is continuous at $x = c$, then so is $|f(x)|$.

(ii) If $f(x)$ and $g(x)$ are discontinuous at $x = c$, then so is $f(x) + g(x)$.

(iii) If $f(x)$ and $g(x)$ are discontinuous at $x = c$ then so is $f(x)g(x)$. (6)

(b) Find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ when

(i) $y = \sin^3\left(\dfrac{2x}{x^2-1}\right)$,

(ii) $y = x^{\cos(2x)}$. (6)

(c) (i) Find the Taylor series expansion of $f(x) = \ln(1+x)$ about $a = 0$ up to the term in x^4.

(ii) Taylor's theorem states that the remainder term in an expansion about $a = 0$ is

$$R_n(x) = \left| f^{(n+1)}(c)\frac{x^{n+1}}{(n+1)!} \right|,$$

where c lies in $(0, x)$. By considering the maximum and minimum values of the remainder term of the series expansion of $f(x) = \ln(1+x)$, show that the expansion converges to $\ln(1+x)$ on the interval $x \in \left(-\dfrac{1}{2}, \dfrac{1}{2}\right)$. (8)

[20]

5. (a) Using the Intermediate Value Theorem, find the root of

$$y = x^4 + x - 1$$

that lies in the interval $x \in (0, 1)$ to 2 decimal place accuracy. (3)

(b) Find the vector equation of the plane containing the three points $A(1, 1, 4)$, $B(2, -5, 4)$ and $C(7, -1, 0)$. (7)

(c) (i) Find the vector equation of the line of intersection of the two planes $\mathbf{r} \cdot (\mathbf{i} + \mathbf{j} + \mathbf{k}) = 5$ and $\mathbf{r} \cdot (4\mathbf{i} + \mathbf{j} + 2\mathbf{k}) = 15$.

(ii) Find the acute angle between the planes. (10)

[20]